

Represent objects in 2 dimensions; discuss and reason about the advantages and disadvantages of different representations

Learning intention

- represent objects in 2 dimensions; discuss and reason about the advantages and disadvantages of different representations.

Teach and model

- deconstructing packaging to identify shapes and nets.
- building objects by interpreting isometric and perspective drawings.
- using isometric and square grid paper to draw views of front, back, side, top and bottom of objects.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.